Context

Currently we have a set of referrals and waiting times projections for England that are frequently used in press releases and published content. While the NI projections have recently been produced, the data for the England projections is now a couple of years old, and we've received requests around if these statements could be updated. Furthermore, with influencing ramping up in Scotland and Wales around their elections, it would be beneficial to produce estimates for these nations too.

Statements to be updated/produced:

By 2030 in England, there are expected to be XX% more urgent cancer referrals each year, up from XXX,000 in 2025 to around X,XXX,000.[1]

More than XXX,000 cancer patients in England are estimated to begin treatment later than they should over the next five years if current performance continues.[2]

[1]Previously calculated based on the difference between urgent suspected cancer referrals from July 2023 to June 2024 compared with the projected number of referrals estimated in July 2028 to June 2029

[2]Previously calculated by modelling the number of patients projected to start treatment on the 62-day pathway, and applying the mean performance figure from the last 6 months of historic data. By calculating the difference between this and how many patients would have started treatment within 62 days if 85% of patients were seen on time, 301,000 are projected to not start treatment within standard from July 2024 to June 2029.

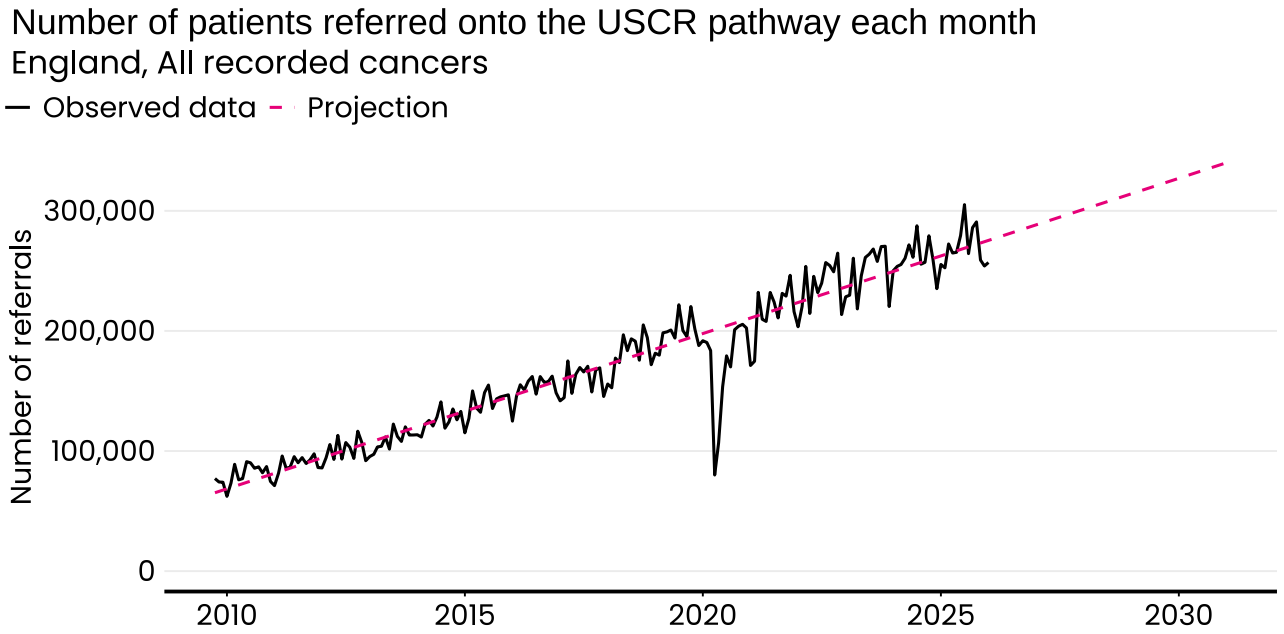
Urgent referrals statement

Data source: `USCR to FOA tab of the SST dashboard`

We think we should preference the FDS statement. If we are to use the urgent referrals statement it shouldn't appear alongside the FDS statement (due to how easily they can confuse).

This was previously calculated based on the difference between urgent suspected cancer referrals from July 2023 to June 2024 compared with the projected number of referrals estimated in July 2028 to June 2030.

By 2030 there are expected to be 23% more urgent cancer referrals each year in England, up from around 3,250,000 in 2025 to around 4,000,000 referrals.[3]



[3]This statement only covers urgent suspected cancer, which is different to the later FDS numbers which have a wider inclusion criteria.

This metric was previously treated as the total number of USC referrals. Technically I think it is slightly less, and is instead the number of USC referrals that have attended their first outpatient appointment. For example, 2022/23 USCR data shows 2,889,406 patients, whereas USC referral numbers list 2,896,111. This is a small difference, and it is reasonable to expect all referrals to have an outpatient appointment during their care.

Faster Diagnosis Standard

Data source: `FDS tab of the SST dashboard`

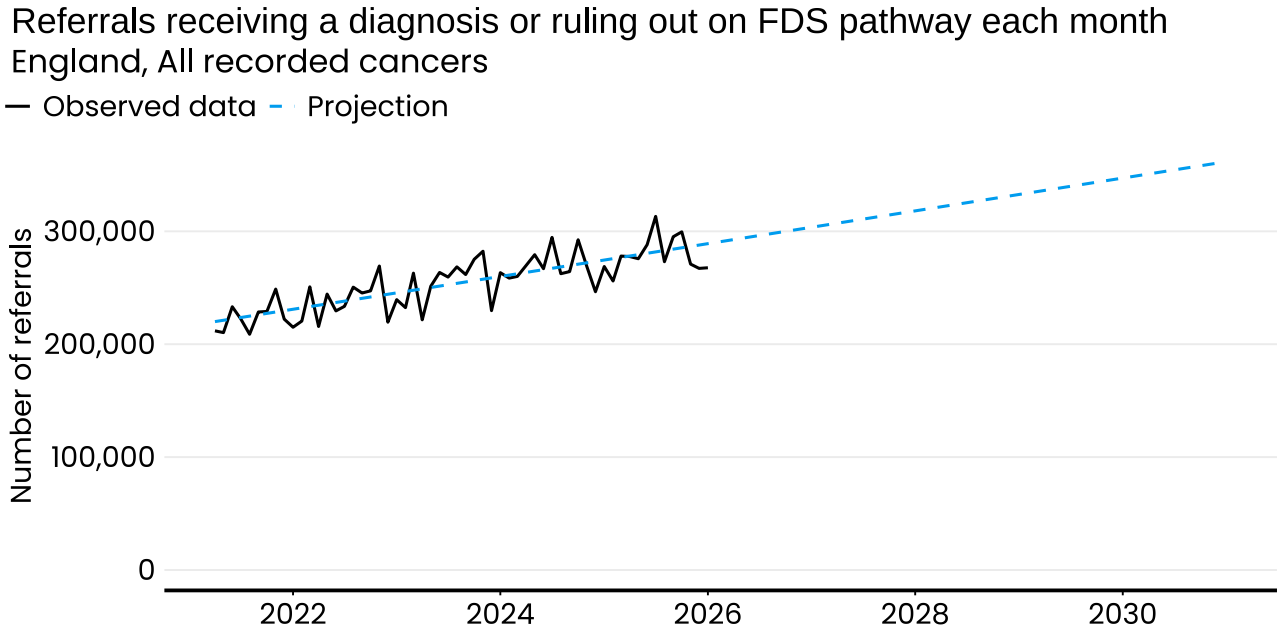
NHS England has the Faster Diagnosis Standard (FDS) which focuses on timely diagnosis. The target is that 80%[^3] of people referred should not wait more than 28 days from referral to finding out whether you have cancer or not. The standard includes urgent suspected cancer referrals, urgent screening referrals and breast symptomatic referrals. It does not include consultant upgrades that is included in the 62 day metric.

By 2030 there are expected to be 26.2% more urgent cancer referrals receiving a diagnosis or having cancer ruled out, each year in England, up from around 3,360,000 in 2025 to around 4,250,000.

12 month average performance (76.2%)

If current FDS performance continues, nearly 719,000 people in England are estimated to wait longer than they should to found out if they have cancer or not over the next five years.

We need to adjust the people wording here as it is referrals not people, and a person could have multiple referrals (approximately one fifth of people have more than 1 referral).



62 day statement

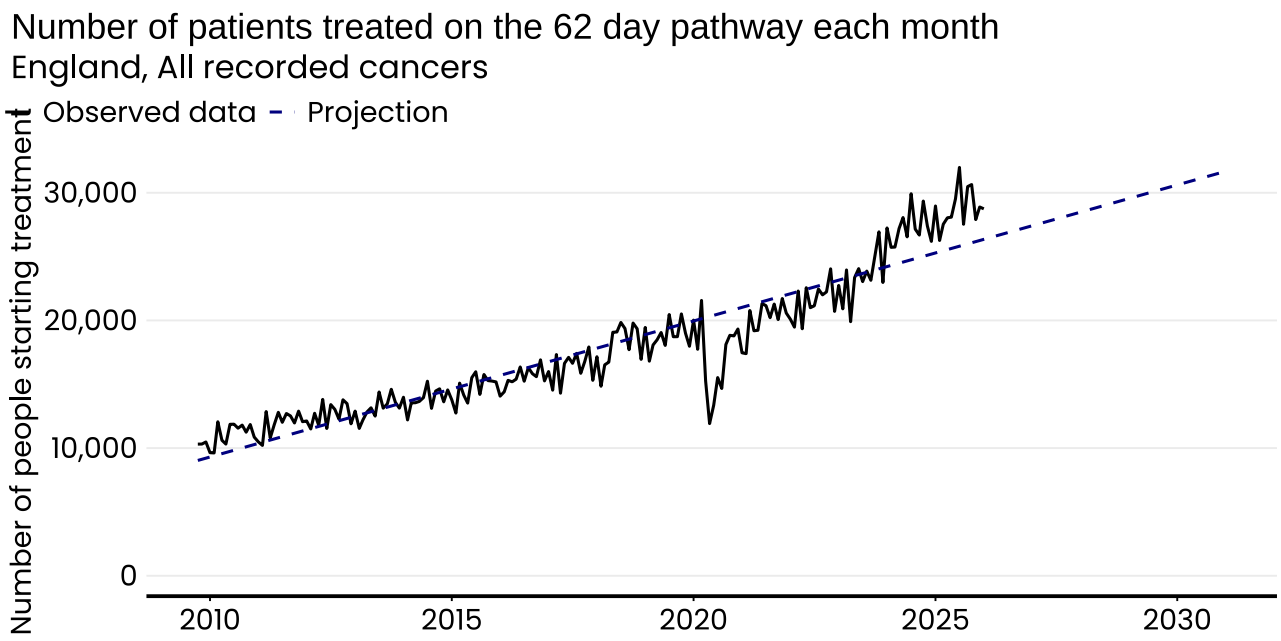
Data source: `SST hub and CWT site (for historical other routes)`

In England, 85% of patients should start their first definitive treatment within 62 days of an urgent suspected cancer referral, breast symptomatic referral, urgent cancer screening programme referral or consultant upgrade.

Previously calculated by modelling the number of patients projected to start treatment on the 62-day pathway that were referred and applying the mean performance figure from the last 6 months of historic data.

We initially did linear regression for this which is used for all the other metrics, however we don't think this really fits the trend in the data, so also ran a poisson model which seems to be much more appropriate.

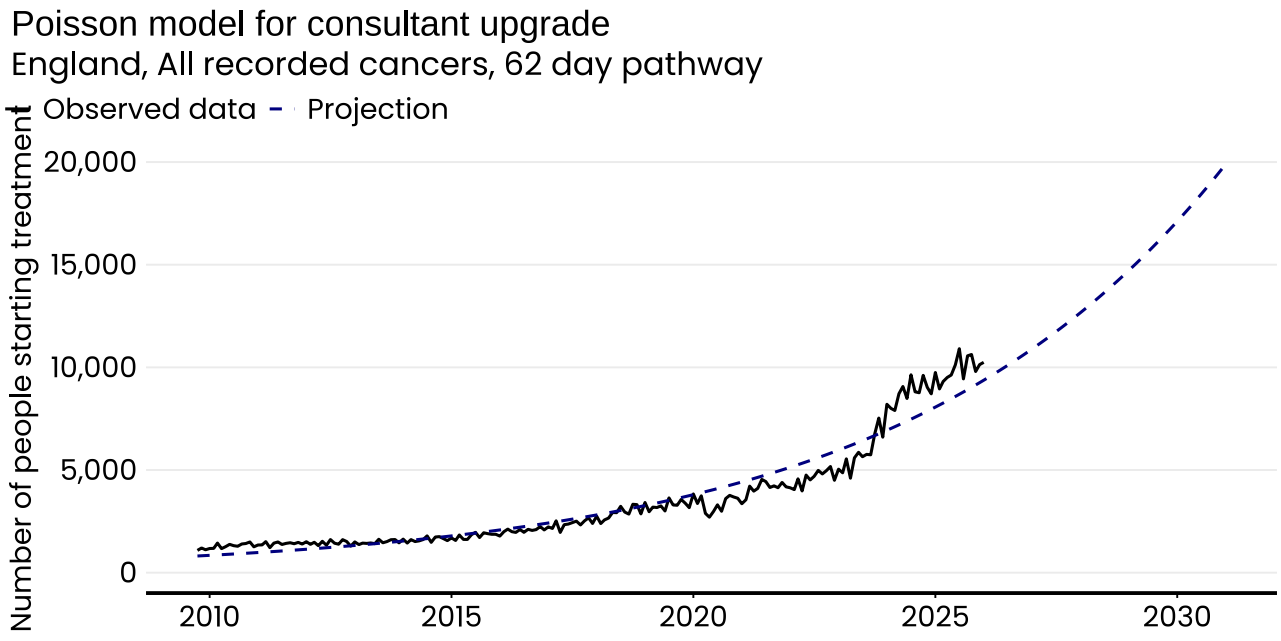
Linear modelling



With the linear regression model, there is a clear sustained increase in number of patients starting treatment after October 2023. If we did use this model, then the final numbers would likely underestimate the actual number of patients of the next 5 years.

Poisson modelling

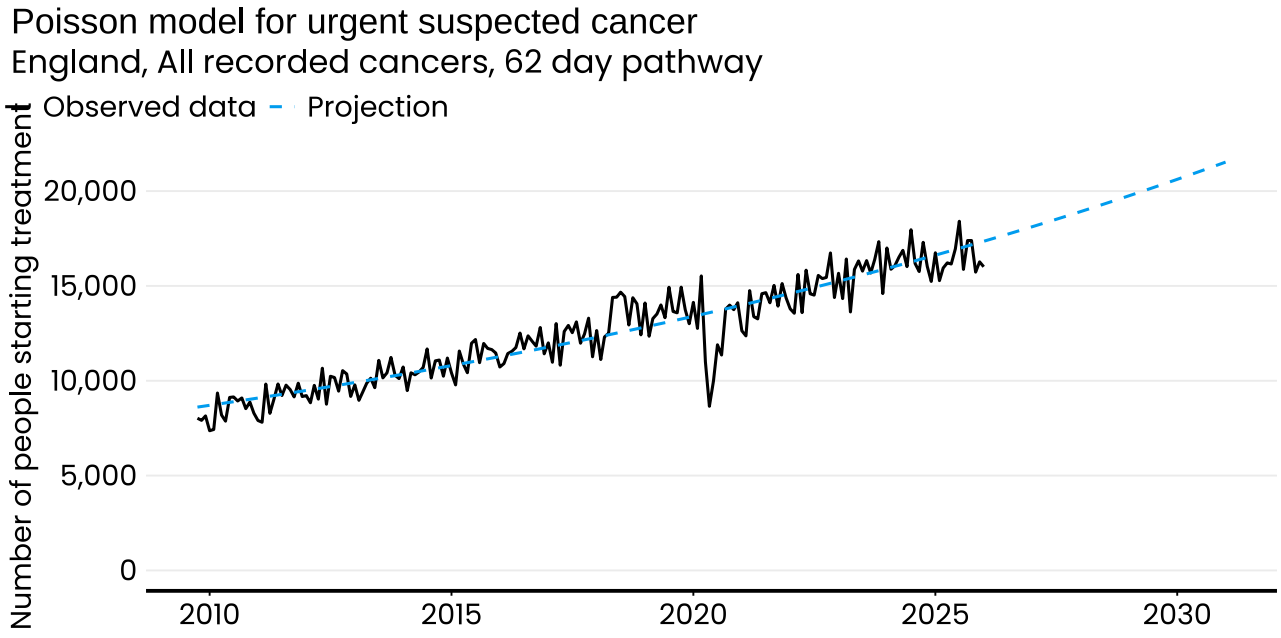
There are different trends present for each referral route, so I have modelled them separately. I have also modelled the three major referral routes together - the breast symptomatic route has not been included in this work as it only goes back to October 2023, but does contain around 100 people each month.



The poisson model for consultant is a much better fit.

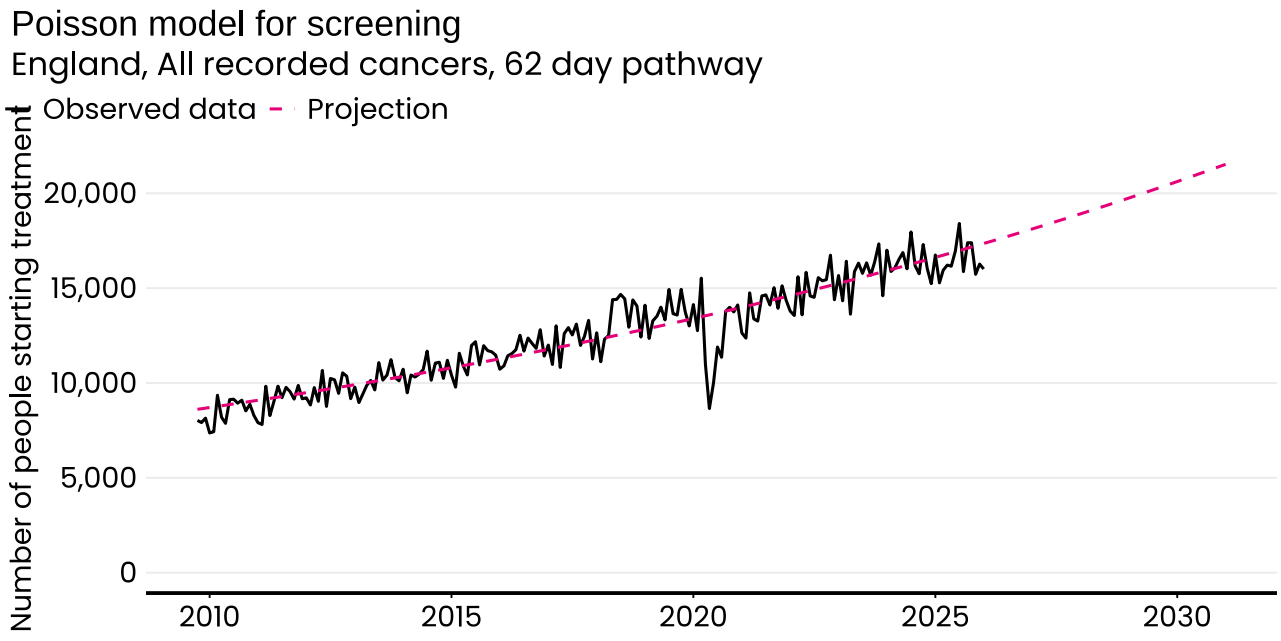
It isn't clear why there was such a large jump in consultant referrals from October 2023, but this is when it became part of the official standard. Data for this referral was available back to 2009 though, when it was far lower. This jump is what is missed by the linear model, and consultant has a different trend to other routes for some reason.

Consultant



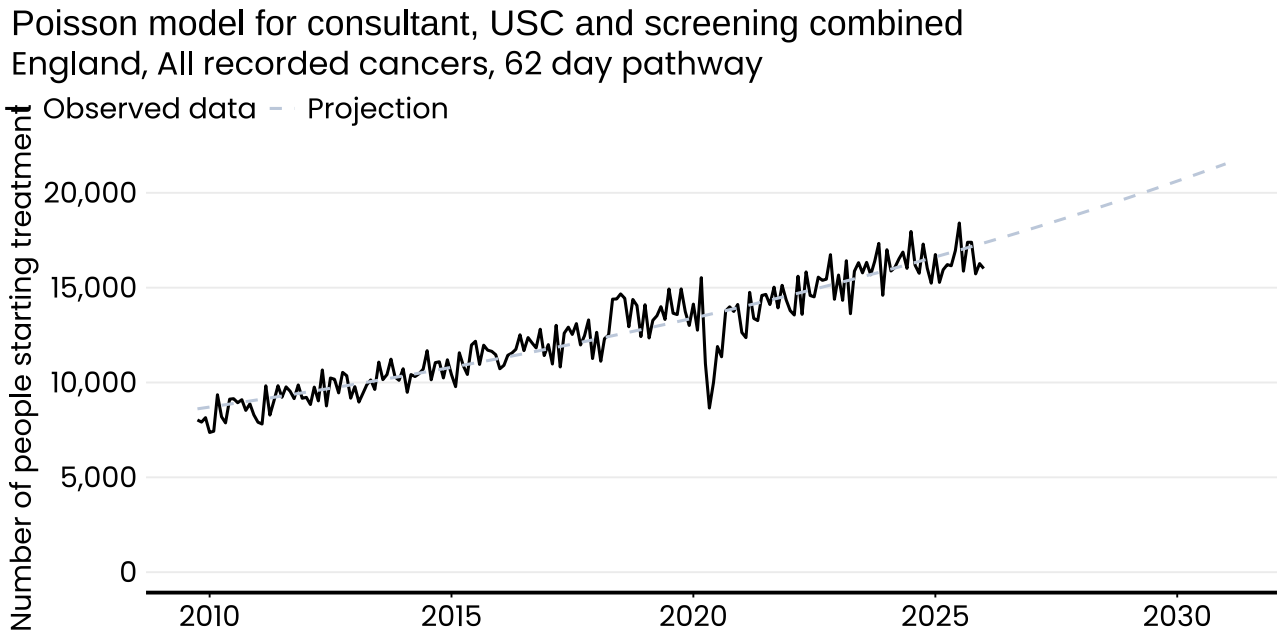
USC has a broadly linear trend, which the poisson model has retained.

USC



Screening seems to be following a linear trend if you consider the whole time series. I don't know what the reasoning was for the gradual increase up to the end of 2017, or what caused it to seemingly reset back down to the linear trend.

Screening

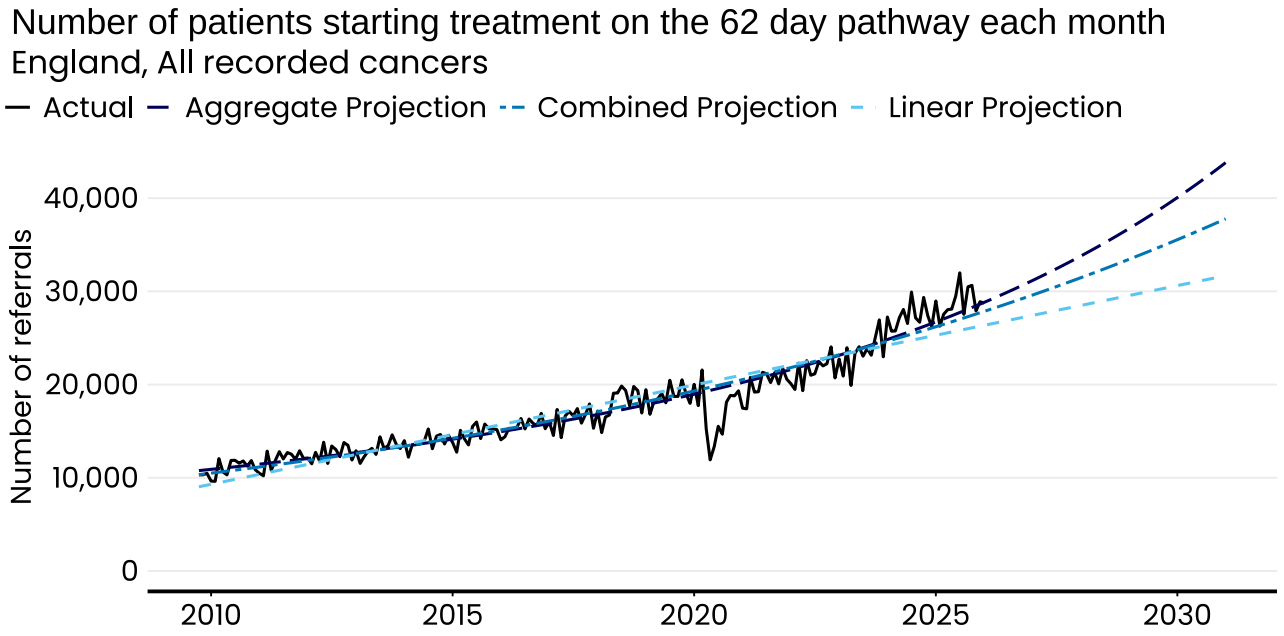


This combined referral route model does a reasonable job of following the data trend. However, given that there are different trends in each of the referral routes, it makes more sense to have three different models, and then sum the results of those models.

Combined

Aggregated 62 day projection

In the following graph you can see the comparison of poisson modelling compared to linear modelling. Given that there are different trends in each of the referral routes, it makes more sense to have three different models, and then sum the results of those models which is the **aggregated** projection on this graph.

12 month average performance (69.2%)

Nearly 337,000 cancer patients in England are estimated to begin treatment later than they should over the next five years if current performance continues.